Population Projections & Demographic Forecasting

Lecture 2: Matrix Projections

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Course overview

- Lecture 1: Population Projections & Cohort Component Method
- Lecture 2: Matrix Projections I
- Lecture 3: Matrix Projections II
- Lecture 4: Demographic Forecasting & Lee-Carter

Small recap

In Lecture 1, we have seen the most employed model for population projections, the cohort component method:

- takes into account age composition of populations
- projection of age groups rather straightforward, except for the youngest and the oldest
- projected population from one interval becomes the baseline for next projection
- several projections can become cumbersome if done one at a time

$\Rightarrow$ we can speed things up with matrix algebra

Introduction

The cohort component method:

- can be compactly written in matrix notation (Bernardelli 1941, Lewis 1942, Leslie 1945, as reported by Smith and Keyfitz 1977)
- rewriting facilitates computer programming and shows the use of matrix algebra for population projections
- simplifies projections over multiple time intervals

Cohort component formulas I

We can write down the female-specific equations for each age group:

$$
\begin{aligned}
N_0(t+5) &= \sum_x N_x(t) b_x \\
N_5(t+5) &= N_0(t)\, s_0 \\
N_{10}(t+5) &= N_5(t)\, s_5 \\
&\vdots \\
N_{45}(t+5) &= N_{40}(t)\, s_{40} \\
N_{50}(t+5) &= N_{45}(t)\, s_{45} \\
&\vdots \\
N_{80}(t+5) &= N_{75}(t)\, s_{75} \\
N_{85+}(t+5) &= \big(N_{80}(t) + N_{85+}(t)\big)\, s_{80}
\end{aligned}
$$

with $s_x = \frac{L_{x+5}}{L_x}$, $s_{80} = \frac{T_{85+}}{T_{80}}$ and $b_x = \frac{1}{1+SRB}\,\frac{L_0}{2\ell_0}\left(F_x + s_x\, F_{x+5}\right)$

Cohort component formulas II

Note the structure of the equations:

$$
\begin{aligned}
N_0(t+5) &= N_{10}(t)b_{10} + \cdots + N_{45}(t)b_{45} \\
N_5(t+5) &= N_0(t)\, s_0 \\
N_{10}(t+5) &= N_5(t)\, s_5 \\
N_{15}(t+5) &= N_{10}(t)\, s_{10} \\
&\vdots \qquad \ddots \\
N_{45}(t+5) &= N_{40}(t)\, s_{40} \\
N_{50}(t+5) &= N_{45}(t)\, s_{45} \\
&\vdots \qquad \ddots \\
N_{80}(t+5) &= N_{75}(t)\, s_{75} \\
N_{85+}(t+5) &= \big(N_{80}(t) + N_{85+}(t)\big)\, s_{8}
\end{aligned}
$$

$\Rightarrow$ this suggests the use of matrix notation

Matrix notation I

Let's rewrite this in matrix form:

$$
\begin{bmatrix}
N_0(t+5) \\
N_5(t+5) \\
N_{10}(t+5) \\
N_{15}(t+5) \\
\vdots \\
N_{45}(t+5) \\
N_{50}(t+5) \\
\vdots \\
N_{80}(t+5) \\
N_{85+}(t+5)
\end{bmatrix}
=
\begin{bmatrix}
0 & 0 & b_{10} & \dots & b_{45} & 0 & \dots & 0 & 0 & 0 \\
s_0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\
0 & s_5 & 0 & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\
0 & 0 & s_{10} & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\
\vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\
0 & 0 & 0 & \dots & s_{45} & 0 & \dots & 0 & 0 & 0 \\
0 & 0 & 0 & \dots & 0 & s_{50} & \dots & 0 & 0 & 0 \\
\vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\
0 & 0 & 0 & \dots & 0 & 0 & \dots & s_{75} & 0 & 0 \\
0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 & s_{80} & s_{80}
\end{bmatrix}
\begin{bmatrix}
N_0(t) \\
N_5(t) \\
N_{10}(t) \\
N_{15}(t) \\
\vdots \\
N_{45}(t) \\
N_{50}(t) \\
\vdots \\
N_{80}(t) \\
N_{85+}(t)
\end{bmatrix}
$$

Matrix notation II

More compactly:

$$\boldsymbol{N}(t+5) = \boldsymbol{L}[t, t+5]\boldsymbol{N}(t) \tag{1}$$

If we can assume that the projection (Leslie) matrix $\boldsymbol{L}$ does not change over future projections, then:

$$\boldsymbol{N}(t+5\times n) = \boldsymbol{L}^{\mathbf{n}}\boldsymbol{N}(t)$$

Matrix projections: Exercise

Exercise

Load the data that we saved yesterday, `Lecture2.Rdata`. Project the female population by one period using the matrix notation in Equation 1. Check your results with the projections that we made in Lecture 1.

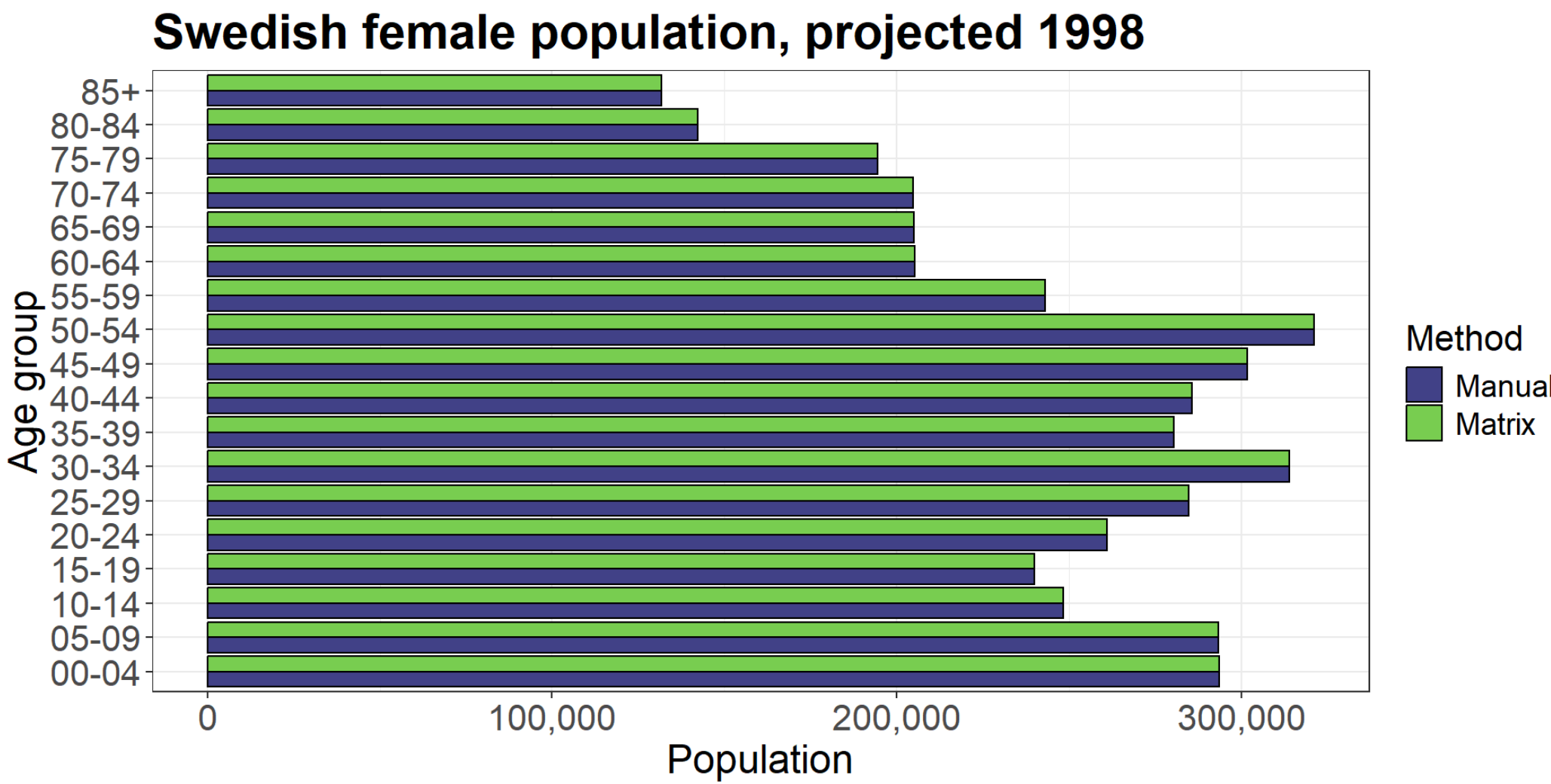
Hint: insert the vectors `sFx` and `bFx` in the matrix $\boldsymbol{L}$ and compute $\boldsymbol{N}(t+5) = \boldsymbol{L}[t, t+5]\boldsymbol{N}(t)$

$$\begin{bmatrix} N_0(t+5) \\ N_5(t+5) \\ N_{10}(t+5) \\ N_{15}(t+5) \\ \vdots \\ N_{45}(t+5) \\ N_{50}(t+5) \\ \vdots \\ N_{80}(t+5) \\ N_{85+}(t+5) \end{bmatrix} = \begin{bmatrix} 0 & 0 & b_{10} & \dots & b_{45} & 0 & \dots & 0 & 0 & 0 \\ s_0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & s_5 & 0 & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & s_{10} & \dots & 0 & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & s_{45} & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & s_{50} & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & s_{75} & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 & s_{80} & s_{80} \end{bmatrix} \begin{bmatrix} N_0(t) \\ N_5(t) \\ N_{10}(t) \\ N_{15}(t) \\ \vdots \\ N_{45}(t) \\ N_{50}(t) \\ \vdots \\ N_{80}(t) \\ N_{85+}(t) \end{bmatrix}$$

One possible solution

```r
### cleaning the workspace
rm(list=ls())
## loading packages
library(tidyverse)
## loading data
load("data/Lecture2.Rdata")
## dimensions
m <- nrow(df.projection)
## extract sx, bx and NFx
sFx <- df.projection$sFx; bFx <- df.projection$bFx; NFx <- df.proje
## change NA to 0 in bFx and remove NA from sFx
bFx[is.na(bFx)] <- 0
sFx <- sFx[!is.na(sFx)]
## create empty Leslie matrix
L <- matrix(0,nrow = m,ncol = m)
## assign bFx and sFx
L[1,] <- bFx
diag(L[-1,]) <- sFx
```

One possible solution



Matrix projections: Exercise II

Exercise

Now that we are sure that the matrix formulation works, let us project $n = 20$ periods ahead. Plot your results in a pyramid showing the starting population, the first and the last projected periods

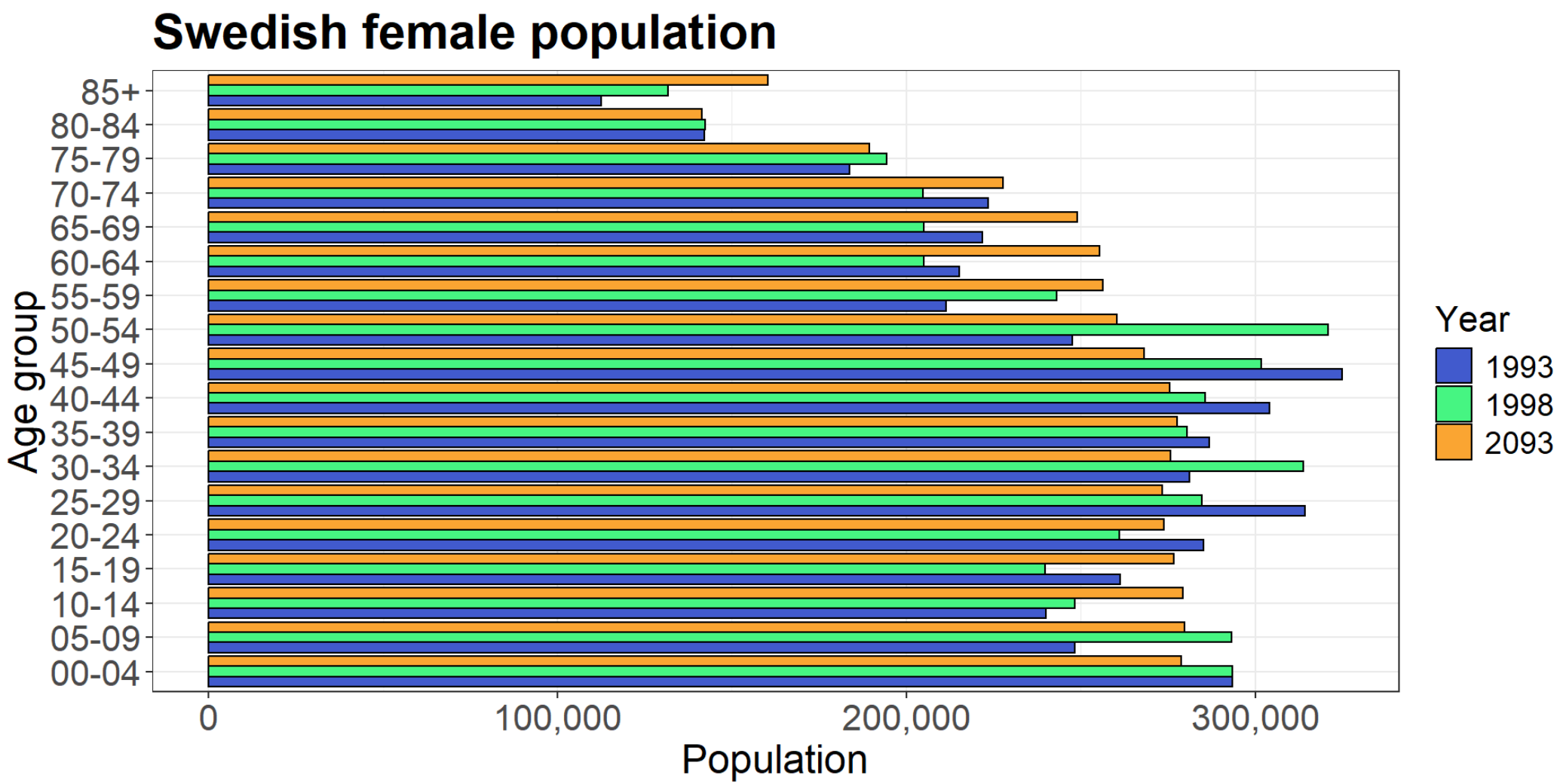
Reminder: $\boldsymbol{N}(t + 5 \times n) = \boldsymbol{L}^{\mathbf{n}}\, \boldsymbol{N}(t)$

Hint: write a function containing a population matrix $\boldsymbol{N}$ with $n + 1$ columns and loop through its columns with matrix multiplication. You can do this recursively (without the need of matrix exponentiation) by updating your baseline population and using the simpler formula

One possible solution

```r
## function to project several periods
pop.proj <- function(x,agegroup,NFx,sFx,bFx,n=1){
  ## number of age groups
  m <- length(x)
  ## create Leslie matrix
  L <- matrix(0,m,m)
  L[1,] <- bFx
  diag(L[-1,]) <- sFx
  L[m,m] <- sFx[m-1]
  ## create population matrix
  N <- matrix(0,m,n+1)
  N[,1] <- NFx
  for (i in 1:n){
    N[,i+1] <- L%*%N[,i]
  }
  out <- cbind(data.frame(x=x,agegroup=agegroup),N)
  return(out)
}
## actual projection
```

One possible solution



Including the male population

- For males, we have seen that we can use (almost) the same formulas as for females
- we can express the equations for both sexes using matrix notation and a block diagonal Leslie matrix:

$$\begin{bmatrix} \boldsymbol{N}^F(t+5) \\ \boldsymbol{N}^M(t+5) \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}^F & 0 \\ \boldsymbol{B}^M & \boldsymbol{L}^M \end{bmatrix} \begin{bmatrix} \boldsymbol{N}^F(t) \\ \boldsymbol{N}^M(t) \end{bmatrix}$$

where

$$\boldsymbol{B}^M = \begin{bmatrix} 0 & 0 & b_{10} & \dots & b_{45} & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix}, \quad \boldsymbol{L}^M = \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ s_0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & s_5 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & s_{10} & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & s_{75} & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & s_{80} & s_{80} \end{bmatrix}$$

Two-sex projections: Exercise

Exercise

Starting again from the data of the previous lecture, write a function to project both sexes contemporaneously. Check if your projection for the first period are the same as those of Lecture 1. Plot a pyramid of the projections after 20 periods.

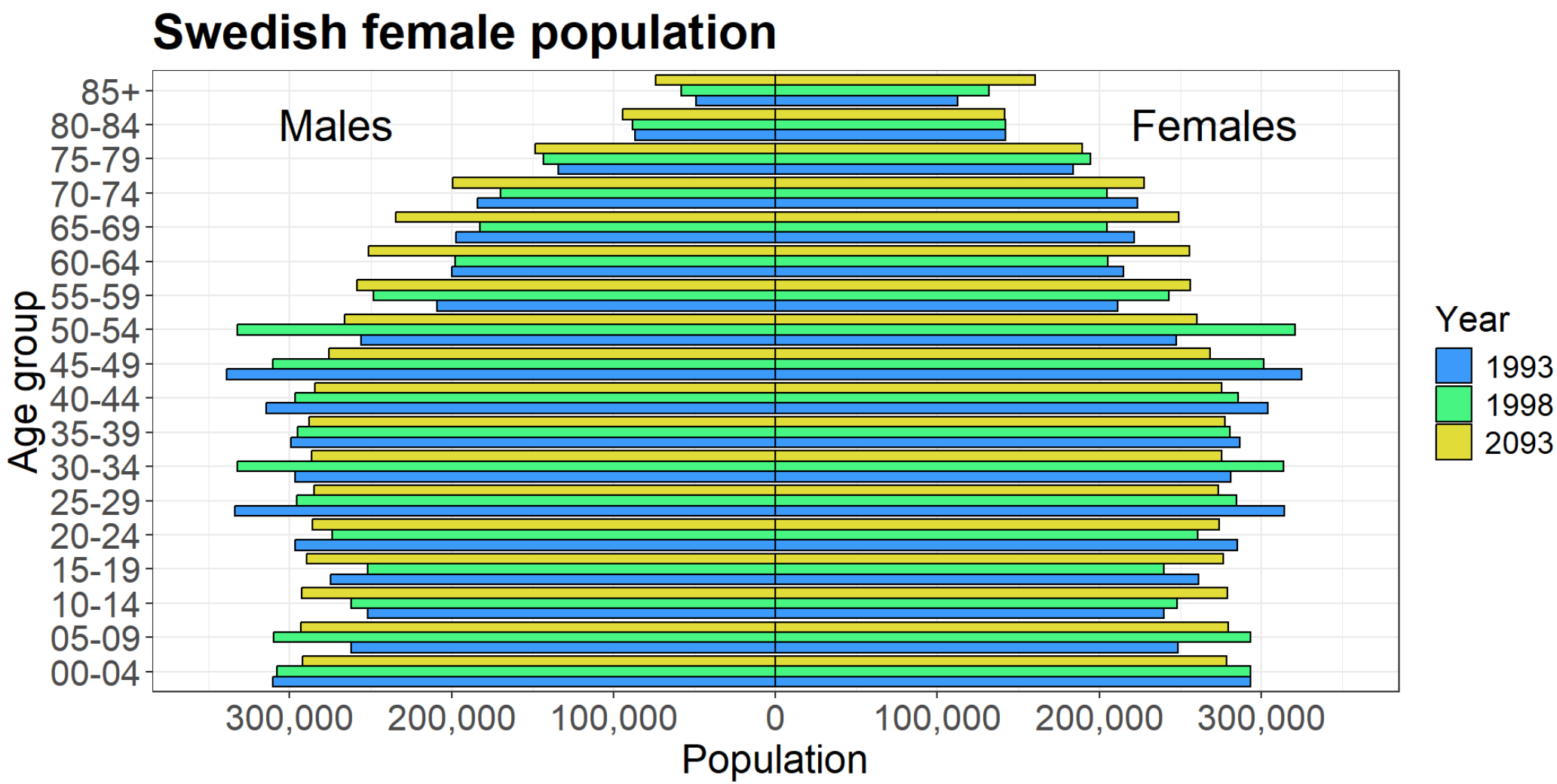
Reminder:

$$\begin{bmatrix} \boldsymbol{N}^F(t+5) \\ \boldsymbol{N}^M(t+5) \end{bmatrix} = \begin{bmatrix} \boldsymbol{L}^F & 0 \\ \boldsymbol{B}^M & \boldsymbol{L}^M \end{bmatrix} \begin{bmatrix} N^F(t) \\ N^M(t) \end{bmatrix}$$

One possible solution

```
sMx <- df.projection$sMx; bMx <- df.projection$bMx; NMx <- df.proje
## similar adjustments to bMx and sMx
bMx[is.na(bMx)] <- 0
sMx <- sMx[!is.na(sMx)]
## female Leslie matrix (derived before)
LF <- L
## male Leslie matrix
BM <- LM <- matrix(0,m,m)
BM[1,] <- bMx
diag(LM[-1,]) <- sMx
LM[m,m] <- sMx[m-1]
## putting male and females together
ZEROS <- diag(0,m)
Lup <- cbind(LF,ZEROS)
Ldown <- cbind(BM,LM)
L <- rbind(Lup,Ldown)
## matrix projection and comparison
Nx <- c(NFx,NMx)
NxF.matrix <- c(L%*%Nx)
```

One possible solution



Some final remarks

- matrix algebra can significantly speed up your population projections
- can incorporate female and male populations within the same setting
- great care is needed to construct the Leslie matrix
- we still have not considered migration, nor time-specific assumptions on future demographic components

$\Rightarrow$ we'll look at these in tomorrow's lecture

Day 2 assignment

Assignment

3. Take the country of interest that you selected for exercise #1 and its population in the last available year. Project its population until the last available date (e.g. 2100) using as inputs the Swedish mortality and fertility patterns that we have used in class, and compare the projected population under this scenario with the one projected by the global providers. What are the main differences that you observe? Briefly comment on the differences between the observed patterns.

4. Take the data for the same population that you selected for exercise #2. However, this time take the data 40 years before the last available date (e.g. if the last available year is 2024, take the data in 1984). Divide the population into 5-year age groups and project the population 40-year ahead for each sex separately using matrix projections. Plot and compare your results with the observed population in the last available data (e.g. in 2024). Are your projections close to the observed population? At what ages do you see larger differences? Briefly provide a motivation for this discrepancy.

References

Bernardelli, H. (1941). Population Waves. *Journal of the Bourma Research Society*, **31**(1), 1–18

Lewis, E.G. (1942). On the generation and growth of a population. *Sankhya*, **6**, 93–96

Leslie, P.H. (1945). On the use of matrices in certain population dynamics. *Biometrika*, **33**, 183–212

Smith, D.P. and Keyfitz, N. (*eds.*) (1977). *Mathematical Demography: Selected Papers*. Berlin: Springer Verlag